Trust Dynamics in Multi-Agent Strategic Interaction: A Simulation Study of Bayesian Reputation Systems in 8-Player Limit Texas Hold'em

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***Abstract*—Reputation systems that infer trustworthiness from observed behavior are widespread in online marketplaces, credit scoring, and social platforms; the transparency paradox—that a perfectly visible cooperative record earns lower returns than a less legible one—remains a hypothesis to test rather than an established fact. We test it in a controlled simulation of 8-player fixed-limit Texas Hold'em in which every agent maintains a categorical posterior over the archetype of every other agent and updates it after each observed action. Holding the game and trust model fixed, we vary only the agents' minds across four architectures: frozen rule-based archetypes, bounded online hill-climbing, large-language-model personality role-players (Claude Haiku 4.5), and the same model augmented with chain-of-thought, per-opponent memory, and self-updated strategy notes. Across five random seeds per condition (twenty, at temperature 0, for the final architecture), the negative Pearson correlation between trust and final stack attenuates monotonically along the ladder $r_{\mathrm{tp}} = -0.752 \pm 0.073 \rightarrow -0.637 \pm 0.125 \rightarrow -0.510 \pm 0.268 \rightarrow +0.062 \pm 0.378$ ($\pm$ = SD across seeds); an unbounded hill-climbing robustness check ($\Delta r_{\mathrm{tp}} = +0.028$) suggests the trap is not an artifact of the archetype bound boxes. The 95% bootstrap interval for the final architecture, $[-0.10, +0.22]$, is centered on zero and excludes the correlations of all earlier architectures, consistent with substantial and possibly complete attenuation, and because hand counts fall from 10 000 per seed in the rule-based phases to 500 and 150 in the LLM phases, the ladder mixes a capability effect with a statistical-power effect. We take this as preliminary evidence that, in our zero-sum testbed, observation-based reputation rewards exploitation under simple agent policies and that reasoning about the reputation system itself substantially attenuates the effect; whether the attenuation is structural or a small-sample artifact is left to a hand-count-matched replication.**

***Index Terms*—Bayesian reputation systems, multi-agent simulation, trust dynamics, game theory, large language models**

I. Introduction

Trust is a fundamental part of human society that is the foundation for human interactions that we have every day. Every business deal, every bank loan, every purchase, and even every relationship is built on trust. This is also reflected by modern-day systems, for example: Amazon shows product reviews, banks check credit scores before lending out loans, and Reddit ranks posts by community voting. Picture this: There are two real estate development firms. The first one is working on their first project; they are new in the market. The second one is an established firm with good reviews and a good history. Now, who would you trust with your life savings to build your dream home? You see, visible reputation systems seem like a win-win. It is easy for you to make the right decision while the trusted agents are rewarded.

Here is the breaking point: this transparency comes with a contradiction. A trusted agent is almost by definition always predictable. When it is right to cooperate, they cooperate, and because everyone knows this in an open environment, the opponent can plan around them. This is what was observed in the real world: the worry is that a perfectly predictable cooperator can be planned around and, in principle, exploited. The empirical record is far from settled, though—the cleanest controlled studies of eBay reputation actually find a *premium* for established sellers rather than a penalty [1, 2]—so we treat this transparency paradox as a hypothesis to test rather than an established fact, of the kind that predictably behaving market-makers also face [3].

We wanted to answer this question but in a controlled environment where we tampered with each of the features, like the rules, the agents, and the trust dynamics between them, and saw for ourselves how they would affect the outcome. While we could draw conclusions from real economic data, we could not run controlled experiments on them; therefore, we settled on running computer simulations. We asked whether reputation systems that infer trust from behaviour ended up rewarding exploitation. Was the system's loss of trust economically optimal? Was the answer more dependent on the design of the trust system or the design of the characters themselves?

Poker turned out to be the perfect environment to try this. Each hand was a set of interactions that mapped directly to real-life interactions—while there was some hidden information (for example, a player's private cards), public actions (bets, calls, and folds) were visible to everyone at the table. Fixed-limit Texas Hold'em was the way to go, where both the bet sizes and action spaces were small enough to make the math tractable but also rich enough for these strategies to emerge. Please note that poker is only the behavioural microscope and not a benchmark to solve.

Specifically, we built a poker simulation using eight players in which each player embodied a distinct playing style. Each agent maintained a perception about the others with a probability distribution to understand whether the opponent is a "tight player who only plays strong hands" or a "loose maniac who bets with a deuce-seven", and so on. After every action, the agents updated their belief systems. Trust was the honesty we expected from an opponent according to our level of confidence in their archetype. We calculated the Pearson

correlation[1] between the economic return earned and the trust coefficient. We varied the nature of the agents, starting relatively simple and becoming more complex. The game and the trust system were the same across phases; only the agents changed. This study could not have been possible without the established work presented in the following section.

II. Background

There are five lines of previous research motivating this study. We summarize them here in the order they influence our design:

- The experimental economic literature on reputation systems in the Internet age: this is the body of research documenting the cost of transparency.
- Computational studies on cooperation in repeated games—including the Prisoner's Dilemma to Axelrod's tournaments.
- The formal treatment of reputation in economics and in computational trust systems—where the object we measure comes from.
- Bayesian opponent modelling in poker—our inference mechanism.
- The still-open question of whether large language models can reason strategically, or only mimic strategy.

Together, these define our core question: does the cost of being trusting depend on either the trust system or on the agents inside it?

A. Online reputation systems

An empirical approach to understanding the economics of online reputation systems has been conducted by Resnick and Zeckhauser [1], who have documented that, based on large-scale data from eBay, feedback is submitted by users greater than fifty percent of the time and is virtually all positive, with sellers receiving negative feedback less than 1% of the time (i.e., evidence of ubiquitous observable reputation and markets rewarding good behavior). Broader surveys of online feedback mechanisms [4] and of the trust and reputation systems built on them [5] describe the same basic architecture that we model here: a public, observation-derived score that other participants condition their behaviour on. The underlying economic problem such systems exist to solve is information asymmetry between parties to a transaction [6], and the mechanism by which an informed party conveys quality to an uninformed one is costly signalling [7]—a framing we return to in our honesty score, which is defined precisely as the reliability of an agent's betting signal. Therefore, the motivation behind our research is a theoretical transparency paradox rather than an empirically verified phenomenon: when the behavior of a cooperator is completely predictable, a sufficiently astute counterpart can plan accordingly and take advantage of the cooperator's actions—hence potentially creating conditions under which visible cooperation could be exploited. Controlled studies using eBay actually point in the opposite direction, i.e., toward a reputation premium associated with established identities [8]—thus making the paradox a hypothesis worth testing versus an empirically demonstrated effect. A similar dynamic is observed in financial-market microstructure, where adverse selection against predictably acting market-makers [3] creates equivalent problems due to differences in information asymmetry. However, because the prior literature does not replicate or alter the population of agents participating in such interactions, it cannot determine whether the paradox would remain even if cooperative agents had sufficient intelligence to prevent being taken advantage of—so a controlled simulation was developed to address that question.

[1]The Pearson correlation coefficient measures the linear relationship between two paired variables on a scale of -1 (perfect negative) to $+1$ (perfect positive); $r = 0$ means no linear relationship, and $|r| \approx 0.5$ is conventionally a moderate effect.

B. Reputation as a formal object

Reputation also has a long formal treatment in game theory, which supplies the vocabulary we use throughout. Kreps & Wilson [9] and Milgrom & Roberts [10] showed that when one player is uncertain about another's *type*, that uncertainty is itself strategically valuable: a player may act against its short-run interest in order to shape the beliefs others hold about it. Fudenberg & Levine [11] sharpened this into a general result about patient players, and Mailath & Samuelson [12] give the modern synthesis. Our setup is a computational instance of exactly this structure—a population of agents holding beliefs over one another's types, with those beliefs feeding back into behaviour—with one deliberate difference: in the classical models the reputation-holder chooses how to invest in its reputation, whereas in Phases 1 and 2 our agents cannot. That asymmetry is the point of the phase ladder, and it is only in Phase 3.1 that agents acquire anything resembling the strategic control over their own reputation that this literature assumes.

C. Cooperation in repeated games

Second, there is a substantial amount of computational research concerning cooperation in repeated games. Prior to discussing cooperation, it is necessary to discuss the Prisoner's Dilemma as it is widely recognized as the paradigmatic example of the conflict between cooperation and exploitation. Each prisoner has two options: (A) Cooperate or (B) Defect. Each prisoner makes their decision simultaneously. There are four possible outcomes based upon the decisions made by each of the prisoners. When both prisoners choose option (A)—i.e., cooperate with one another—each receives a moderate reward. When one prisoner chooses option (A) (cooperates) and the other selects option (B) (defects), the cooperator incurs a loss while the defector receives the largest reward and pays no penalty. When both prisoners select option (B), i.e., defect, neither receives any reward. Mutual cooperation provides a reward of $+2$ to both players; mutual defection provides a reward of 0 to both players; and unilaterally defecting against a cooperator provides a reward of $+3$ to the defector and a loss of -1 to the cooperator. In terms of rational self-interest, each prisoner has sufficient incentive to defect since defecting always yields greater benefits than cooperating, regardless of the choice made by the other prisoner; Nevertheless, mutually defective

behavior is inferior in comparison to mutually cooperative behavior.

Axelrod investigated this problem extensively through his Prisoner's Dilemma Tournaments [13], and Nowak developed his five rules for the evolutionary development of cooperation [14], having earlier shown with May that spatial structure alone can sustain cooperation without any reciprocity mechanism at all [15]. The strand of that literature closest to our own concern is *indirect* reciprocity, in which agents condition their behaviour not on what a partner did to them but on what a partner did to third parties, as summarised by a public image score [16, 17]. Our trust posterior is an indirect-reciprocity mechanism in this precise sense: an observer updates its beliefs about an agent from actions taken against everyone at the table, not merely against the observer. The experimental literature on human trust behaviour—the investment game of Berg et al. [18] and the costly-punishment results of Fehr & Gächter [19]—supplies the two behaviours we hand-built into **Mirror** and **Judge** respectively. Axelrod demonstrated that reciprocity-based strategies such as Tit-for-Tat are robustly effective in one-to-one situations. Agents that cooperate by default and punish defectors on a tit-for-tat basis tend to perform effectively over time against all types of competitors. This lends credence to the notion that cooperation can represent a viable strategic alternative. However, Axelrod's conclusions are predicated upon two assumptions which fail to apply in our context. Firstly, all interactions are now no longer one-to-one but instead involve eight separate agents. As a result, the negative impact associated with aggressive behavior on behalf of a single individual is diluted across seven other competing agents versus being focused solely upon one opponent. Secondly, the set of actions available for an agent to take extends far beyond simply cooperating or defecting. Poker allows an agent to fold, call, raise, bluff[2] and slow-play—all of which add additional degrees-of-freedom for an agent seeking to extract value from their opponent(s). Our best approximation to Tit-for-Tat is **Mirror**—which ranks 6th out of 8 in terms of Phase 1 stacks—suggesting that Axelrod's conclusions do not necessarily translate well to multi-agent environments that contain richer action-spaces; although we interpret this as suggestive rather than definitive.

D. Bayesian opponent modelling in poker

Third, there is Bayesian opponent modeling in poker. Initially this included Billings et al.'s Loki and Poki bots [20] and subsequently formalized by Southey et al. [21] as Bayesian-type inference— followed by Ganzfried and Sandholm's [22] further generalization toward safe-exploitation of an opponent. Most recently, Carnegie Mellon University's Libratus and Pluribus [23, 24] reached super-human performance levels in heads-up and six-player no-limit Texas Hold'em. While that body of work differs from our own objectives—specifically seeking optimal or safely exploitative behavior to win—we borrow their type-inference machinery and utilize poker as a controlled environment for examining how trust dynamics evolve under conditions of correct, transparent, and collective inference amongst many agents. Our poker engine is viewed as a microscope—not as an adversary.

[2]A *bluff* is a bet made with a weak hand intended to make opponents fold stronger hands; its inverse, a *value bet*, is a bet with a strong hand intended to be called by weaker hands. *Fold equity* is the value an aggressor captures when opponents fold without reaching showdown.

E. Language models as strategic agents

Finally, we treat this last point as it relates to two of our phases, because these phases employ large language models. An increasing amount of work concerns how strategically large language models—used as agents in competitive and observable settings—behave. Current benchmarks for negotiation and game playing, such as AgentBench [25], and Park et al.'s study on generative agents [26], suggest that there is still much to learn about this issue. Phase 3 and phase 3.1 are both designed to examine the difference between how large language model-based agents actually behave and what they represent by way of textual strategies.

Two quantitative tools recur throughout the analysis: the Pearson correlation coefficient, which we compute between an archetype's mean trust score and its final stack—a value we call r_{tp}—and Bayesian updating, by which each agent maintains a categorical posterior over the 8 archetypes for every other player and revises it on every action it observes. Both are standard; Supplementary Section S1 states them formally. The next section describes the smallest controlled setup combining all five lines of prior work.

III. Methodology

Throughout every phase of this study, the only variable is the agent's mind. The underlying scaffolding includes three components: an 8-player version of fixed-limit Texas Hold'em; categorical belief about each player's archetype maintained by each player and updated after every round; and one single trust score computed based upon belief.

In phase 1, agents operate under pre-defined rule-based archetypes. In phase 2, agents allow their rules to evolve through limited hill climbing. In phase 3, agents replace their rules with a language model that represents themselves playing an archetype. In phase 3.1, agents add explicit reasoning—chain-of-thought, per-opponent memory, and self-generated strategy documentation. Since neither the environment nor the trust model ever changes throughout this experiment, differences in outcomes can be solely due to variations in agent capabilities.

First we describe the common environment, followed by descriptions of the eight archetypes (rules), belief updates, the experimental design, and the metrics; we close with a detailed description of the four agent implementations, one per phase.

A. Game environment

The overall game setting remains constant for all phases of this study. We will utilize a fixed-limit version of eight player Texas Hold'em, where the small blind is 1, the large blind is 2, the small bet (2) can be made pre-flop or on the first round of cards (the "flop"), the large bet (4) can be made on the second round of cards ("turn") and third round of cards

("river"). There is a cap of four-bet allowed per round. Each player starts a run with a chip count of 200 in their respective seat; stacks then carry across hands, and when a player's chips run out, he/she receives a rebuy of another 200. Side pots[3] are used to determine who wins a showdown; a side pot is created by players contributing money at their individual betting level. Side pots are made and awarded depending on whether a player played chips into the respective pot and the remainder goes to the player who comes first from the dealer, going in clockwise direction.

B. The eight agent archetypes

The eight archetypes (cooperative vs. exploitative; static vs. adaptable; based on opponent vs. based on action type) span the axes identified by Game Theory and Behavioral Economics as "load bearing." The policies for each agent are tables of probability distributions over actions per round given their hand strength bucket. The static types have fixed parameters during all simulations. Each of the eight types also has an *honesty* score $h = 1 - P(\text{bluff} \mid \text{weak})$, which is used as a weighting factor in computing the trust scores (Eq. 3). In addition, the eight type names are set in bold throughout the text so the reader can track which player is being discussed at any point. The full per-archetype policy bounds—bet, bluff, and fold probabilities and the adaptation rule for each of the eight types—are given in Supplementary Table S2.

Oracle | *Baseline* | honesty 0.67
The honest "baseline" from which other players deviate. This player plays an approximate game-theoretic optimal mix and is used as the model by which all adaptive players are evaluated.

Sentinel | *Honest (tight-aggressive, TAG)* | honesty 0.92
The truthful player. Folds most hands and plays aggressively when it enters. It signals that its bet is strong, while signalling that its check is weak.

Firestorm | *Maniac (loose-aggressive)* | honesty 0.38
A cheater. Bets and raises very frequently, whatever the strength of its hand. Primarily wins due to opponents folding rather than winning at showdown.

Wall | *Calling station* | honesty 0.96
The player who never folds and never bluffs. The most-trusted player because it is so predictable; but in Phases 1 through 3 it is also the weakest economically for the same reason.

Phantom | *Deceiver* | honesty 0.48
An early aggressive bluffer that folds to resistance. Described as being placed in a vague region of the archetype space.

Predator | *Opponent classifier* | honesty 0.79
Based upon the posterior beliefs about an opponent's type, this player changes strategy to exploit them if they have greater than a 0.6 probability of being a non-TAG player. If the opponent has less than a 0.6 chance of being a non-TAG player, this player returns to its base-case behaviour of playing like the Sentinel.

Mirror | *Tit-for-Tat* | honesty 0.91
Takes the most-active opponent's behaviours and blends them with its own policy. In theory this represents the poker version of the Iterated Prisoner's Dilemma (IPD) Tit-for-Tat strategy; however, in practice it has been shown to work poorly in multi-player games.

Judge | *Punishes bluffing* | honesty 0.92
Maintains a record of how each opponent has interacted with it; after five instances of an opponent bluffing, this player takes revenge on that opponent. Models retaliation based upon perceived cheating or "bad faith."

[3]When a player goes all-in for fewer chips than the largest contributor, the engine creates a *side pot* for the excess so that the short-stacked player can only win up to the amount they contributed; deeper-stacked players contest the remainder among themselves.

C. Categorical posterior over types

Every agent maintains a length-8 categorical distribution over the archetype of every other seat. When agent i sees seat j perform action a in round r, the agent updates the distribution as

$$p_{i\to j}(t) \;\propto\; \left[(1-\varepsilon)\, L(a \mid t, r) + \varepsilon/|\mathcal{T}|\right] p^{\text{prior}}_{i\to j}(t), \tag{1}$$

where $L(a \mid t, r)$ is a precomputed likelihood that a type-t agent emits action a in round r, $\mathcal{T}$ is the set of eight archetypes, and $\varepsilon = 0.05$ is a noise floor preventing any type from being fully eliminated. We characterize Eq. 1 as Bayes-flavoured but not strictly Bayesian. The noise floor is encoded into the likelihood function (a uniform contamination model). The likelihood function is computed from the analytic policy for each archetype, so it is well-specified. Equation 2 models the between-hand flattening of the posteriors towards a uniform prior. Each component is raised to the power λ and the result renormalized—an exponential tempering step:

$$p_{i\to j}(t) \;\leftarrow\; \frac{p_{i\to j}(t)^{\lambda}}{\sum_{t'\in\mathcal{T}} p_{i\to j}(t')^{\lambda}}, \qquad \lambda = 0.95, \tag{2}$$

Under this update the log-odds between any two types decay by the factor λ per hand, providing an effective memory of about 20 hands (approximately $1/(1-\lambda)$). The trust score, our main dependent variable, is defined as

$$\text{trust}_{i\to j} \;=\; \sum_{t\in\mathcal{T}} p_{i\to j}(t)\, h(t), \tag{3}$$

where $h(t) = 1 - P(\text{bluff} \mid \text{weak})$ is the archetype's honesty score. As previously stated, because the state-switching archetypes (**Predator**, **Judge**, and **Mirror**) have different characteristics that define their level of honesty across states, $h(t)$ represents each archetype's nominal, canonical level of honesty—that is, the expected level of honesty implied by their respective mean bluff rates—and not their actual level of honesty in a specific state. As such, trust will represent the expected level of honesty that an observer anticipates based upon the type it believes the agent to be. There are many reasonable ways to define the trust score; here we have chosen this simple form, and we report results under this definition throughout. In addition, in our limitations section we note that because our definition of the trust–profit relationship is mechanically tied to the above definition of the trust score, alternative definitions could produce different relationships. *Three archetypes (**Sentinel**, **Mirror** in its default state, **Judge** in its cooperative state) are intentionally given near-identical mean parameters to demonstrate a controlled case of identifiability among indistinguishable types.* Although **Sentinel** and cooperative **Judge** are byte-identical in the four statistics the likelihood model reads (their policy bound boxes differ; Supplementary Table S2), **Mirror** varies only slightly from **Sentinel** and also dynamically adjusts toward its target at runtime; as a result, the realized behavior of **Mirror** differs from that of **Sentinel** (see Supplementary Table S4). This is a design choice, not a finding.

D. Experimental design

All four phases use the same five random seeds (42, 137, 256, 512, 1024), and all phases used the same eight archetypes; the canonical Phase 3.1 dataset extends these to twenty seeds (the original five plus fifteen drawn once from a fixed generator and frozen before any outcome was observed). However, there was a huge difference in how many hands were played within each phase that can account for why

r_{tp} differed so much among the different phases. Specifically, Phases 1, 2, and 2* each played 10 000 hands per seed; Phase 3 played 500 hands per seed (total 43 943 LLM calls, $33.10); and Phase 3.1 played 150 hands per seed across its twenty seeds (∼48 000 calls, ∼$68). It is well understood that when comparing results from multiple phases of data collection using a similar model but at very different levels of sampling intensity, it will create both a capability effect as well as a statistical-power effect. Therefore, the differences in r_{tp} observed between the various phases also reflect a mix of these two effects. We discuss this issue explicitly in the discussion section (Limitations), and any future hand-count matched replications of the data collected here would provide a way to isolate one or both of these issues.

Sampling temperature. The Phase 3 dataset was generated at the Anthropic SDK's default temperature of 1.0. The canonical Phase 3.1 dataset reported here was generated with temperature pinned at 0.0, matching the released codebase; an earlier five-seed Phase 3.1 exploration at temperature 1.0 produced a statistically consistent estimate ($r_{\mathrm{tp}} = -0.094 \pm 0.301$). The remaining Phase 3 temperature caveat is described in the Limitations.

E. Metrics

The key measure was the Pearson correlation coefficient r_{tp} per seed for the average degree of trust (final hand) and the number of chips in each player's stack at the end of all games (average number of chips in the stacks of all players) across the eight different types of players (8 pairs of measurements in total, one per type). Since these eight points represent a predefined, engineered set—with two or three archetypes clustering at nearly equal trust values—each per-seed r_{tp} should be viewed as a descriptive correlation over that design rather than a sample correlation drawn from a population.

We also present six auxiliary measures, including the index of exploitation of trust (TEI)[4], sensitivity to context (CS)[5], adaptation by the opponent (OA)[6], non-stationarity (NS)[7], unpredictability in strategy (SU), and awareness of manipulation of trust (TMA).

Awareness of manipulation of trust (TMA) is calculated as the correlation between a player's rolling trust delta and its rolling aggression delta. Positive values in this case can show patterns of building up trust during the initial stages of cooperation followed by increasing aggression after building trust (which we call "trust farming" in captions and further discussions).

These six auxiliary metrics were selected post-hoc rather than pre-registered. We report the metrics above descriptively, with no inferential tests; an individual value should therefore be interpreted as indicative rather than as a statement of statistical significance. Reporting six metrics without correcting for multiplicity would, if these were inferential claims, invite exactly the error that false-discovery-rate procedures exist to control [27]; choosing the metrics after seeing the data compounds it [28]. We therefore make no significance claim on any of the six.

Throughout, a reported ± value denotes the (population) standard deviation of r_{tp} across that phase's seeds. Confidence Intervals for the headline Pearson r_{tp} correlations are provided in two ways: a t-interval based upon the Student-t distribution with $n-1$ degrees of freedom, where n is that phase's seed count (for the five-seed phases, $T_{\mathrm{crit}} = 2.776$; for the twenty-seed Phase 3.1, 2.093), and a non-parametric percentile bootstrap [29] with 10 000 iterations. Although a five-seed phase draws from at most $\binom{2n-1}{n} = 126$ distinct bootstrap combinations of players, thus ensuring dense coverage of the sample space via the 10 000 iterations, it does not provide greater statistical power than what is available from the five actual observations. Per-seed intervals on an individual r_{tp} use the Fisher z-transformation [30] on the $n = 8$ archetype pairs underlying that correlation.

F. The four agent architectures in detail

The phases share every component described above and differ only in how each agent turns the current game state into an action. We describe each in turn; Supplementary Table S2 gives the numerical policy that each phase starts from.

a) Phase 1 — frozen rules.: All eight agents are "lookup tables" which map the current betting round to the agent's hand-strength bucket (as determined by the Monte Carlo evaluator), and to a probability distribution over fold, check, call, bet, and raise. The agent randomly selects its action based on the resulting probability distribution. Nothing about the table is changed throughout a run. Therefore, Phase 1 represents the best possible baseline for measuring each subsequent phase. Although the three adaptive archetypes (**Predator**, **Mirror**, and **Judge**) continually draw upon live state (opponent posterior or a grievance ledger), the rules which consume that state are also frozen.

b) Phase 2 — bounded online optimisation.: Every agent uses a hill climber to adjust its policy on a per-cycle basis. When the cycle starts it introduces a random (round, metric) adjustment of size $\delta = 0.03$ to its own policy, plays the entire cycle, and retains the adjustment only if its net profits during the cycle were greater than previous cycles; otherwise it is reverted. Since the agent is only able to introduce adjustments within a "bound box" drawn around its canonical parameters (see Supplementary Table S2), a **Sentinel** may improve its value betting but is unable to move into the bluff-everything area of **Firestorm**. Therefore Phase 2 attempts to answer a narrowly focused question: can we achieve escape from the trap using local, reward-greedy optimisation *within* a personality?

c) Phase 2 — unbounded optimisation.:* This robustness variation removes the bound box from consideration entirely, increases the step size to $\delta = 0.15$, and decreases the number of rounds used to evaluate policy improvements from 200 hands down to 50. Thus Phase 2* tests whether the archetype enclosures—instead of the trust dynamics—form the barrier which holds the trap.

d) Phase 3 — LLM personality role-players.: All decisions are made by Anthropic's Claude Haiku 4.5. The agent's personality specification serves as input to the system prompt; hole card details, board details, pot details, betting history and stack sizes serve as input to the user message; and the model response is converted back into one of the permitted actions. In addition to this lack of memory across hands and absence of explicit reasoning process, the model simply role-plays each personality one decision at a time at the SDK's default temperature of 1.0.

e) Phase 3.1 — LLM with reasoning scaffolding.: In Phase 3.1 the same model and personality prompts as Phase 3 are used. However, three additional scaffolds are introduced. A two-sentence chain-of-thought field requires that the model state a reason prior to taking action; a per-opponent memory is refreshed every ten hands, summarizing how each opponent has played previously; and a self-authored strategy note is rewritten every twenty-five hands and allows an agent to maintain a long-term plan from hand to hand. These scaffolds allow an agent to reason about how it is being viewed and act to reduce the predictability of that view and thus the reputation system that rewards that view—and this is the sole intervention that moves r_{tp} to zero and slightly past it.

IV. Results

We present results in order of agent capability: Phase 1 (frozen rule-based agents) establishes the baseline trap; Phase 2 (bounded hill-climbing) tests whether numerical adaptation can soften it; Phase 2* (unbounded hill-climbing) tests whether the bounds were the binding constraint; Phase 3 (LLM personality role-players) replaces the rule-based policies with a language model under a personality spec; and

[4] *TEI* (Trust Exploitation Index): an agent's average non-showdown winnings per hand—wins generated via fold equity rather than at showdown.

[5] *CS* (Contextual Sensitivity): the absolute correlation between an opponent's past aggression and the agent's response action.

[6] *OA* (Opponent Adaptation): the variance of an agent's aggression rate across different opponents.

[7] *NS* (Non-Stationarity): how much an agent's action distribution changes over the run, comparing 500-hand windows against its overall distribution.

Archetype	Trust	Stack	σ	Rebuys	Rank P3	Rank P3.1	Δ
mirror	0.748	280	81	0.0	2	1	+1
oracle	0.618	238	112	0.1	3	2	+1
phantom	0.631	236	70	0.1	1	3	-2
predator	0.770	235	111	0.1	6	4	+2
sentinel	0.759	234	96	0.0	7	5	+2
wall	0.937	217	109	0.6	8	6	+2
judge	0.760	203	80	0.0	4	7	-3
firestorm	0.449	196	65	0.4	5	8	-3

TABLE I

PHASE 3.1 ECONOMIC OUTCOMES BY ARCHETYPE (TWENTY-SEED TEMPERATURE-0 REPLICATION, 150 HANDS PER SEED). RANKS ARE DERIVED FROM FINAL-STACK ORDERING. FIRESTORM, THE DOMINANT EARNER OF EVERY RULE-BASED PHASE, FINISHES LAST; WALL, THE MOST-TRUSTED ARCHETYPE, ENDS ABOVE ITS 200-CHIP BUY-IN WITH REBUYS COLLAPSING FROM 9.4 PER SEED (PHASE 1) TO 0.6.

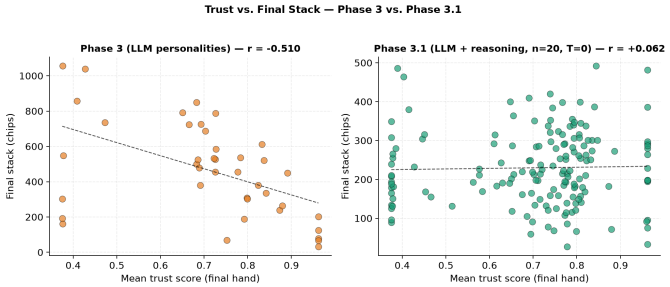


Fig. 1. **Trust versus final stack in the LLM phases.** (**Left**) Phase 3: the negative trust–stack relationship persists under LLM personality role-play ($r_{\mathrm{tp}} = -0.510$). (**Right**) Phase 3.1 (twenty seeds, temperature 0): with reasoning scaffolding the relationship dissolves ($r_{\mathrm{tp}} = +0.062$); trust no longer predicts final stack. Each point is one (archetype, seed) observation.

Phase 3.1 (LLM with reasoning scaffolding) augments that with chain-of-thought, per-opponent memory, and self-updated strategy notes. The four-tier headline ladder $-0.752 \rightarrow -0.637 \rightarrow -0.510 \rightarrow +0.062$ (Phase 2^*—a separate, unbounded-optimization robustness check on Phase 2—is excluded from this sequential reporting) frames the remainder of this section, which reports per-phase numbers, figures, and illustrative hands. Table I previews the endpoint of that ladder—the final per-archetype economic ordering under Phase 3.1—which the per-phase narrative below then builds toward.

A. Phase 1: trust as a liability

Phase 1 produces $r_{\mathrm{tp}} = -0.752 \pm 0.073$ across the five canonical seeds, with per-seed values $\{-0.774, -0.608, -0.792, -0.812, -0.776\}$. **Wall** (the most-trusted archetype, honesty 0.96) has the smallest mean stack and approximately 9.4 rebuys per seed. **Firestorm** (the least-trusted, honesty 0.38) has the largest mean stack and zero rebuys. Of the pots **Firestorm** wins, 87.1% are walkovers[8]; its showdown win rate is only 38.5%. **Firestorm** is not a strong poker player on the merits; it wins because it is hard to read. This same negative trust–stack relationship recurs in the LLM phases: it persists under personality-only role-play (Fig. 1, left, Phase 3) and only dissolves once reasoning is added (right, Phase 3.1).

The mechanism is also visible in a single representative hand. (**Firestorm** happens to hold a genuine hand here rather than running a bluff; the hand directly illustrates **Wall**'s calling-station leak.) Supplementary Box S1 reproduces hand #67 from seed 42: **Wall** is dealt 2♠ 5♣ (hand rank[9] 6,749, the worst hand at the table) and calls every street of a 3-bet[10] pot, paying 35 chips into a hand it had no chance to win. The illustrative hand we use here is drawn from the Phase 3 dataset so that we can show the trap holding even under a language-model implementation of **Wall**; the same dynamic occurs more frequently in Phase 1.

a) Classification accuracy.: The **Predator** archetype, which reads its posterior to choose exploitation targets, reliably classifies only 2 of 7 opponents above the prespecified 0.60 confidence threshold after 1,000 hands—**Wall** ($p \approx 1.0$) and **Firestorm** ($p \approx 0.82$)—with **Phantom** only occasionally crossing the threshold ($p \approx 0.60$ in some seeds). The remaining four opponents fall below the threshold on every seed. Three of these (**Sentinel**, **Mirror** in its default state, **Judge** in its cooperative state) have near-identical mean parameters under the spec (**Sentinel** and **Judge** are byte-identical in those parameters; **Mirror** is only minimally different and blends at runtime), so the posterior over the cluster saturates at high uncertainty—up to $\log_2(3) \approx 1.58$ bits when all three are treated as indistinguishable, and at least $\log_2(2) = 1$ bit for the byte-identical **Sentinel/Judge** pair—a direct consequence of our design choice rather than an emergent finding.

B. Phase 2: bounded hill-climbing softens but does not close the trap

Phase 2 generates $r_{\mathrm{tp}} = -0.637 \pm 0.125$ with per-seed values $\{-0.759, -0.424, -0.719, -0.717, -0.564\}$. The direction of movement from Phase 1 ($\Delta r_{\mathrm{tp}} = +0.116$) agrees across all five seeds (directional movements of $+0.015$, $+0.184$, $+0.073$, $+0.095$, $+0.212$). The Opponent Adaptation metric also remains constant at OA $= 0.0003$ in both Phase 1 and Phase 2. As such, while Phase 2 optimises aggregate rewards and can alter parameter settings to fit within a bounding box, there is no way to construct a per-opponent strategy since the hill-climber does not have slots available for per-opponent parameters. Agents are shifted approximately 0.3 units of L_1[11] in parameter space over 10,000 hands of play, so the bounded agents remain close to their initial locations. In this regard, the softening in Phase 2 is genuine; however, the mechanism by which it occurs is localised to the agents trimming off the least-effective components of an otherwise stationary policy without introducing anything fundamentally different. Supplementary Fig. S3 compares the bounded and unbounded per-seed correlations directly.

C. Phase 2^*: unbounded hill-climbing fails the same way

Phase 2^* yields $r_{\mathrm{tp}} = -0.609 \pm 0.221$ with per-seed values $\{-0.354, -0.700, -0.344, -0.887, -0.759\}$. The mean increase in r_{tp} from Phase 2 is $+0.028$, well within the standard deviation associated with the seed-to-seed variation. With respect to movement in parameter space, the mean per-agent L_1 drift is 3.4 (i.e., eleven times more than in the bounded version); yet the trap persists (Supplementary Fig. S2 resolves the drift per archetype).

The explanation for why this occurs becomes apparent on examining the dispersal in cluster-spread space. The averaged pairwise L_1 distance between all eight agents in 36-dimensional parameter space expands from an initial 5.82 to ≥ 7.5 on every seed; the mean of the final-to-initial cluster-spread ratios is 1.324. While the agents are moving eleven times harder than in the bounded case, they are not converging on a common Nash-like profile [31]—the natural prediction if one treats eight independent profit-maximisers in a zero-sum game [32] as a system that ought to settle into equilibrium. They diverge

[8] A *walkover* (or "uncontested pot") is a hand won without showdown—every other player folded before the cards had to be shown. It is the bread-and-butter outcome for aggressive players who win by making opponents fold rather than by holding the best hand.

[9] *Hand rank* is taken from standard 7-card poker equity tables: rank 1 is the strongest possible hand and rank 7,462 is the weakest. Rank $\leq$ 1,000 is roughly "top 15% of hands."

[10] *3-bet* and *4-bet* refer to successive re-raises. The big blind counts as the first bet; a raise is the 2-bet; a re-raise is the 3-bet; another re-raise is the 4-bet. Under our four-bet cap, no further raises are allowed in the same round.

[11] The L_1 *distance* between two parameter vectors is the sum of absolute differences across all 36 (round, action) slots. Larger L_1 means the agent has moved further from its starting parameters.

instead. We stress that this is a negative result about *this optimizer at this budget*, not a proof that no equilibrium exists; a population of independently hill-climbing agents is a poor equilibrium-finder, a point made more generally in the multi-agent learning literature on sequential social dilemmas [33].

Firestorm's mean stack continues to remain $\sim 6\times$ that of the next closest archetype; **Wall** still incurs ~ 28 rebuys per seed. The economic ordering does not change. We credit the lack of convergence to three structural causes that we expect would survive any specific optimizer choice—non-stationary feedback from the lagging trust posterior, the simultaneous actions of all eight agents, and axis-aligned coordinate descent in 36 dimensions—and we elaborate in the Discussion.

D. Phase 3: language models alone do not break the trap

Phase 3 generates $r_{\text{tp}} = -0.510 \pm 0.268$ with per-seed values $\{-0.884, -0.525, -0.171, -0.712, -0.259\}$. This is a change from Phase 2 ($\Delta r_{\text{tp}} = +0.127$) of similar magnitude to the Phase 1 $\rightarrow$ Phase 2 step. The variance of r_{tp} among seeds is roughly double that of Phase 2 (0.268 vs. 0.125). Only two of six dimension-specific behavioral targets are achieved; three of the metrics move *backward* relative to Phases 1 and 2 (Supplementary Table S4 reports per-archetype VPIP[12], PFR[13], and AF[14] fingerprints). Among the secondary metrics, CS drops from 0.142 to 0.076, NS collapses to 0, and SU[15] falls from 1.88 to 1.19 bits.

Specifications describing personality characteristics inform an LLM *what* to do, but not *how* to reason about doing so. Absent supporting information about reasoning, the LLMs in our sample collapse onto a more generic, stereotyped canonical interpretation of each archetype, and thus become *more* classifiable than the explicitly stochastic rule-based versions. The primary reason for the softening between Phase 2 and Phase 3 is therefore output variability arising from sampling at temperature 1.0.

E. Phase 3.1: reasoning scaffolding attenuates the trap

Phase 3.1, in its canonical twenty-seed form, generates $r_{\text{tp}} = +0.062 \pm 0.378$, with ten of the twenty seeds producing a positive per-seed correlation. The directional shift from Phase 3 ($\Delta r_{\text{tp}} = +0.572$) is larger than that of any previous phase, and at $n = 20$ seeds the 95% bootstrap interval, $[-0.10, +0.22]$, is about a third narrower than the corresponding five-seed bootstrap interval and centered on zero: it excludes negative correlations of the magnitude reported in every earlier phase while remaining consistent with either full neutralization or a small residual effect of either sign. Individual seeds remain uninformative in isolation—given $n = 8$ archetype pairs per seed, the Fisher-z interval for a per-seed $r_{\text{tp}} = +0.06$ spans roughly $[-0.67, +0.73]$—so the claim rests on the ensemble, not on any single seed. A temperature-matched comparison isolates the scaffold effect from the sampling change: at temperature 1.0 and five seeds, Phase 3 gives $r_{\text{tp}} = -0.510$ and Phase 3.1 gives -0.094 ($\Delta r_{\text{tp}} = +0.416$), so the attenuation is present before temperature is pinned and is not an artifact of it; the temperature-0 twenty-seed run reproduces it at higher power. All statistics and exhibits in this subsection derive from this twenty-seed temperature-0 dataset, with one exception: the boxed illustrative hand is drawn from the original five-seed exploration, whose hand-level transcripts remain the source of the story hands.

Wall's mean stack rises from $\sim$100–200 chips in Phases 1–3 to 217—above its 200-chip buy-in for the first time—and its rebuy count collapses from the 9.4 per seed it needed in Phase 3 to 0.6. **Firestorm**, the top earner of every rule-based phase, finishes last at 196. Fig. 2 shows the rank change for every archetype. The right panel of Fig. 1 (introduced earlier) shows the trust–stack scatter under Phase 3.1: the negative slope visible in the Phase 3 panel is gone. The full per-archetype economic ordering is in Table I, shown at the start of this section.

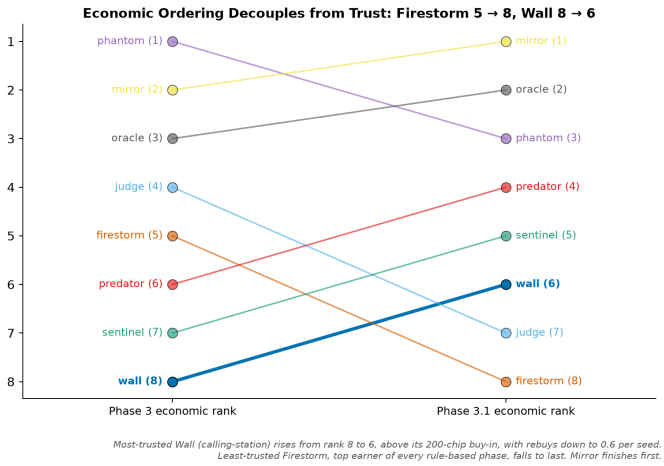


Fig. 2. **Economic ordering decouples from trust between Phase 3 and Phase 3.1. Firestorm**, the dominant earner of every rule-based phase, falls from rank 5 in Phase 3 to rank 8 of 8; **Wall** (trust 0.94, highest of any archetype) rises from rank 8 in Phase 3, on 9.4 rebuys per seed, to rank 6 above its buy-in on 0.6; **Mirror**, the reciprocity archetype, finishes first.

Strategic Unpredictability rises from 1.19 to 1.60 bits, recovering past 1.5 for the first time in the LLM phases (Phases 1–2 sat at 1.88). Trust Manipulation Awareness averages $+0.19$, with seven of eight archetypes showing weakly positive TMA; **Judge** and **Predator** show the strongest signals at $+0.30$ and $+0.28$, and every magnitude is far below the values observed in the temperature-1.0 exploration. For honest archetypes a positive TMA warrants skepticism: reasoning-enhanced value-betting causes trust and aggression to rise together without any deception, exaggerating apparent "farming." Per-archetype TMA values are in Supplementary Table S3; Supplementary Fig. S1 plots them.

The mechanism underlying these values is best shown in a single hand. Supplementary Box S2 reproduces hand #146 of the original five-seed Phase 3.1 exploration: **Wall**, holding $K\heartsuit Q\heartsuit$ (rank 872), calls a flop continuation bet[16] and a turn barrel from **Firestorm**, reads **Firestorm**'s river check as a tell[17], places a value bet on the river, and wins 32 chips. Canonical **Wall** never makes a river bet anywhere in the Phase 1, Phase 2, or Phase 3 datasets; conversely, with chain-of-thought reasoning added to its decision-making during Phase 3.1, **Wall** does.

F. A first scaffold ablation: removing chain-of-thought

Phase 3.1 deploys three scaffolds jointly, so the attenuation cannot be attributed to any one of them without separating them. We report one leave-one-out contrast, chosen because it maps directly onto the paper's central claim: the *no-CoT* arm removes chain-of-thought reasoning while retaining both information scaffolds (per-opponent memory and adaptive notes), at the same five seeds, 150 hands and

[12] *VPIP* (voluntarily put in pot) is the percentage of hands in which a player voluntarily put chips into the pot, excluding the forced blinds. High VPIP = loose; low VPIP = tight.

[13] *PFR* (preflop raise rate) is the percentage of hands in which a player raised before the flop. PFR $\leq$ VPIP by definition.

[14] *Aggression factor (AF)* is the ratio (bets + raises)/calls. AF > 2 is aggressive; AF < 1 is passive.

[15] *Strategic Unpredictability* (SU) is the Shannon entropy [34] in bits over each agent's action distribution. Higher SU = harder to predict.

[16] A *continuation bet* (*c-bet*) is a bet made on the flop by the player who took the lead in betting preflop, regardless of whether the flop helped that player's hand. It is the most common bluff line in modern poker.

[17] A *tell* is a behavioral signal that gives information about an opponent's hand. **Wall** reads **Firestorm**'s river check as a tell because the canonical **Firestorm** policy bets every street with a real hand; a river check therefore signals that the bluff line has been abandoned.

temperature 0 as the canonical run. It yields $r_{\mathrm{tp}} = -0.217 \pm 0.418$ ($n = 5$; 95% bootstrap CI $[-0.54, +0.11]$)—roughly midway between the full-scaffold $+0.062$ and the no-scaffold Phase 3 value of -0.510. Removing the reasoning scaffold while leaving the information scaffolds intact therefore restores a substantial part of the trap, which is what the reasoning-not-optimization reading predicts. At $n = 5$ the interval is wide and overlaps both neighbours, so we read this as directional support rather than a decisive decomposition. The complementary *no-memory* and *no-notes* arms were attempted in the same batch but lost to an API quota exhaustion mid-run; their seeds are excluded rather than reported, because the failures fell in the closing hands that set the final stacks. A complete three-arm ablation remains outstanding.

V. Discussion

The title to this research has been intentionally unsettling; in every agent configuration short of full reasoning scaffolding, being "trusted" represented a fiscal cost rather than a fiscal advantage, and no amount of numerical optimization removed that cost. Only agents able to reason about the reputation system itself—Phase 3.1—escaped it.

The sections that follow establish how the problem develops, why any form of numerical optimization cannot resolve it, and how the use of logical reasoning can keep the problem from occurring, but they also provide insight into what limitations exist when doing so with a small-seed study (five seeds per phase; twenty for Phase 3.1).

A. The mechanism behind the trap

The trust trap visible in Phase 1 admits a clean mechanistic explanation that the Phase 3.1 results then refine. Trust, as we measure it, is the expected honesty of an opponent under the posterior; a high trust score signals to opponents that the agent is most likely an archetype that only bets with strong hands. For **Wall** (honesty 0.96) and **Sentinel** (honesty 0.92), this belief is *correct*, and that is precisely why it is costly: when opponents know an agent only bets with strong hands, they fold against its bets (denying value from those strong hands) and bet aggressively against its checks (exploiting the information that a check signals weakness). The agent's honesty becomes a signal that opponents read and trade against. A low trust score, conversely, creates strategic ambiguity, and **Firestorm** uses that ambiguity to dominate the table. The phenomenon is an instance of the *transparency paradox* introduced earlier—that predictable cooperation can be exploited—which our zero-sum testbed shows can arise under simple agent policies, even though the empirical reputation record does not establish it for real markets.

The mechanism by which **Firestorm** converts ambiguity into chips deserves comment because it differs structurally from the canonical Iterated Prisoner's Dilemma result. In pairwise IPD, Always-Defect is outperformed by Tit-for-Tat because a defector faces concentrated retaliation. In our 8-player setting, **Firestorm**'s aggression cost is diffused across seven simultaneous opponents, so the same strategy that would lose heads-up wins at the full table. Our Tit-for-Tat analogue (**Mirror**) finishes sixth of eight in Phase 1's economic ordering—suggestive but not definitive evidence that the IPD result fails to transfer to multi-agent settings with richer action spaces. We flag this as a direction for follow-up rather than a conclusion, since a single hand-crafted poker analogue is not a clean test of the IPD literature.

We believe it worthwhile to dwell on why this trap is so stable. The loop is self-reinforcing: every honest action taken by an agent provides an informative sample which sharpens the posterior of its opponents and thus makes the next exploitation more precise. Therefore honesty not only costs an agent once but pays for the very classification that will be turned against it as well. If viewed through the lens of signalling theory, trust is a perfectly truthful signal, and in a competitive zero-sum environment a perfectly truthful signal regarding cooperation will provide adversaries with exactly what they require to fold out their value bets and to exploit the checks of honest agents. As such, **Firestorm**'s advantage is simply the opposite side of this coin; by acting almost independent of its hand strength it sends an uninformative signal, and since an uninformative signal cannot be used against it, **Firestorm** exploits this uninformative signal in favor of itself. Therefore, we feel that the trap is less about honesty itself and more about the predictive nature created by honest signals.

B. What numerical optimization can and cannot do

The next three phases of the experiment narrow the space of possible explanations for why the trap is hard to escape. We begin with a structural caveat: by construction, three of our eight archetypes (**Sentinel**, **Mirror**'s default, **Judge**'s cooperative state) share near-identical mean parameters (**Sentinel** and **Judge** are byte-identical in those parameters; **Mirror** is only minimally different and blends at runtime), and the Bayesian posterior cannot reliably distinguish them regardless of algorithmic sophistication. Classification accuracy in that sub-space is bounded above by chance—$1/3$ for three indistinguishable types, $1/2$ for the byte-identical pair. This is a definitional point about *our* archetype space, not an information- theoretic discovery about reputation systems in general; the empirical question of how rapidly classification accuracy degrades as inter-archetype distance shrinks toward zero would require experimentally varying that distance and is outside the scope of the present paper. Real-world monitoring systems—fraud detection, content moderation, credit scoring—face an analogous failure mode for moderately-behaved actors with overlapping profiles, but the analogy is suggestive rather than implied by our setup.

Within that caveat, the Phase 2 and Phase 2* results argue against two specific hypotheses about the trap's origin. Phase 2 (bounded hill-climbing) softens r_{tp} by $+0.116$ while OA stays pinned at 0.0003: aggregate-reward optimization can move agent parameters but cannot, by construction, produce per-opponent strategy because the hill-climber has no per-opponent parameter slot. Phase 2* (unbounded hill-climbing with $11\times$ greater parameter drift) softens r_{tp} by an additional $+0.028$ and produces a population that diverges rather than converges in 36-dimensional parameter space—evidence against the hypothesis that the archetype- shaped bound boxes are the binding constraint, though the sub-experiment varies three optimizer parameters at once (bounds removed, step size raised $5\times$, evaluation window shrunk $4\times$) and therefore does not cleanly isolate the parameter-space hypothesis. The non-convergence has three plausible structural causes that we expect would survive any specific optimizer choice: the trust posterior is non-stationary inside each agent's optimization loop (with $\lambda = 0.95$ the effective memory is $1/(1-\lambda) = 20$ hands, so beliefs take several such windows—on the order of 70 hands—to fully re-converge after a strategy change, longer than the 50-hand evaluation window), all eight agents climb simultaneously so each trial perturbation is evaluated against a moving joint behavior, and axis-aligned coordinate descent in 36 dimensions cannot discover joint optima even with larger steps. A CMA-ES or REINFORCE optimizer with $\sim$50 000 cycles would search more efficiently, but the first two causes would survive any optimization substrate that respects the same observation contract. Supplementary Fig. S4 tracks the resulting dispersion in parameter space.

Phase 3 (LLM personality role-players) softens r_{tp} by a further $+0.127$, but the diagnostic metrics suggest the mechanism is not categorically different from Phase 2's: behavioral-dimension targets fail in three of four directions, with the LLMs collapsing onto a more stereotyped interpretation of each archetype than the explicitly stochastic rule-based versions. The personality specs tell the agents *what* to do but not *how* to reason; absent reasoning support the LLMs role-play the spec without spontaneously inventing opponent-conditional, time-varying, or unpredictable strategy. The Phase 3 softening may therefore be produced by output stochasticity at sampling temperature 1.0 rather than by a structural change in agent behavior.

When combined, the results of Phases 2 and 2* clearly indicate that trap behavior is a characteristic of the observation/exploitation loop, not some specific choice of parameters. The local, reward-greedy optimizer can only nudge an agent toward neighboring policies, which are quickly reclassified and re-exploited by the fixed and shared trust model; there exists no escape route in policy space that

greedy steps can follow, due to the fact that the very gradient along which the agent is climbing is simultaneously defined by opposing agents. That is why increasing the step size disperses the population rather than causing convergence—each agent is chasing targets whose simultaneous adaptation by other agents causes them to move. While numerical optimization can refine a policy, it cannot reason its way past being modeled, and being modeled is precisely the issue here.

C. What reasoning adds, and what we can and cannot claim outside poker

The Phase 3.1 result is consistent with an interpretation in which agent reasoning about the reputation system itself attenuates the trust–profit correlation. The chain-of-thought prompt allows a one-step lookahead; the per-opponent memory gives that reasoning concrete data to condition on; the adaptive strategy notes let the agent update its prior on what is working against which opponent. Whether all three are required we cannot yet claim: only the no-CoT arm of the leave-one-out ablation completed, and while its direction supports the reading above, the two information scaffolds remain untested. The **Wall** hand reproduced in Supplementary Box S2 illustrates the mechanism concretely in a single instance: **Wall** reads **Firestorm**'s river check as a tell, recognizes that K-Q high beats **Firestorm**'s bluffing range, and bets for value—a sequence that is impossible for the Phase 1 **Wall** and unreliable for the Phase 3 **Wall**. A distributional analysis across all Phase 3.1 hands, rather than a single illustrative case, is the natural follow-up.

As opposed to providing more information for an agent to utilize in Phase 3.1, it is how that information is utilized that changes. Both a hill climber and a personality-only language model operate at first order: they either optimize or role-play their own policy. The agent operating with reasoning scaffolding operates at second order: it represents how it is being perceived and acts to destroy the predictability generated by its reputation.

The **Wall** hand in Supplementary Box S2 is a miniature example of exactly this shift; an agent does not only execute "calling station," it infers from **Firestorm**'s out-of-character river check that the board now favors it and departs its script to take value.

The capability which reduces the severity of the trap is meta-cognitive rather than informational—a significantly different prescription from "give the cooperator more features about its counterparties."

These mechanisms suggest hypotheses for real-world reputation systems, but we emphasize the limit of the analogy: our testbed is a zero-sum game and the proposed exploitation mechanism (predictable cooperation extracted via fold equity) does not directly map to positive-sum marketplaces, where the buyer's payoff is from the good rather than from out-thinking the seller. With that caveat, three implications follow as conjectures for cross-domain follow-up. *Behavioral reputation alone may be insufficient* without external enforcement, because observation-based trust can create perverse incentives that reward exploitative strategies. *Classification systems may have blind spots* for moderately-behaved actors with overlapping profiles, mirroring our identifiability sub-space. And *transparency itself may be costly* when it makes cooperative strategies too easy to model and exploit. Each of these would require direct validation in a positive-sum domain to claim more than analogical inference.

Considering all four phases of experimentation, the four-tier ladder presents a single picture as to where the solution for the trap exists. The trap becomes maximized when the rules are frozen (Phase 1); tuning the rules numerically (Phases 2 & 2^*) minimally reduces it; replacing the rules with a fluent but non-reasoning language model (Phase 3) significantly reduces it, primarily through increased output noise; and finally only when the agent has been provided mechanisms to reason about the reputation system (Phase 3.1) does correlation approach zero.

Therefore the remedy, if our reading is correct, is not better policy or richer feature sets but higher-order reasoning—a conclusion that would have gone invisible without holding the game and trust model fixed while varying only the agent's mind.

We consider this monotone ladder the paper's central qualitative result, and the exact magnitude of each rung the part most in need of replication.

D. Limitations

Three clusters of limitations remain after the revisions above.

Statistical power gaps. The headline correlations are computed across $n = 5$ random seeds ($n = 20$ for Phase 3.1). The most common test is simply a single-sample t-test comparing the average r_{tp} value to zero (df $= n - 1$; $T_{\text{crit}} = 2.776$ at five seeds, 2.093 at twenty); the test's p-value therefore depends upon the between-seed variance, not some fixed critical correlation value, and we do not view r_{tp} as if it were itself a correlation over the seed-level points. The headline Phase 3.1 result has a 95% bootstrap interval $[-0.10, +0.22]$; at twenty seeds this excludes residual negative correlations of the magnitude seen in any earlier phase, but still cannot separate exact neutralization from a small residual effect of either sign. Hand counts also differ across phases by an order of magnitude (10 000 hands/seed in Phases 1–2^*, 500 in Phase 3, 150 in Phase 3.1), which mixes a capability effect with a statistical-power effect in the cross-phase ladder. The six secondary metrics (TEI, CS, OA, NS, SU, TMA) were chosen post hoc and are unadjusted for multiple comparisons. The 150-hand $n = 20$ replication has now been run and is the canonical Phase 3.1 dataset reported here; the ∼$230 500-hand-horizon variant, which would additionally remove the hand-count asymmetry against Phase 3, remains open.

Reproducibility and replication gaps. The canonical Phase 3 dataset was generated under the Anthropic SDK's default sampling temperature of 1.0, while the released codebase now pins temperature to 0.0; the Phase 3 headline numbers are therefore not exactly reproducible from the current code without manual reversion. The canonical Phase 3.1 dataset was generated at temperature 0.0 and does not carry this caveat. Phase 3 and 3.1 also use a single model family (Claude Haiku 4.5); a multi-model replication (Gemini Flash, GPT-4o-mini, or a comparable alternative) is needed to demonstrate that the attenuation is not model-specific. Finally, the Phase 3.1 three-scaffold deployment (chain-of-thought, memory, adaptive notes) is only partially separated: we report the no-CoT arm, but the no-memory and no-notes arms were lost to an API quota exhaustion and remain outstanding, so the relative contribution of the two information scaffolds is untested.

Design-choice gaps. Several of our methodological choices are load-bearing in ways that future work should disambiguate. The identifiability "floor" is a stipulated property of our archetype space (three archetypes share mean parameters), not an emergent finding; the honesty score $h(t) = 1 - P(\text{bluff} \mid \text{weak})$ is one of several plausible functional forms (a value-bet-weighted product form is another) and the trust–profit correlation is mechanically anchored on this choice; the rebuy economics underweight deep chip losses (**Wall**'s 9.4 mean rebuys per seed in Phase 1 represent ∼1 880 chips of true loss not visible in the final-stack metric); the Phase 2^* sub- experiment varies three optimizer parameters at once; and Limit Hold'em is zero-sum among players, while eBay marketplaces, social networks, and credit systems are positive-sum over trades, so direct cross-domain inference is rhetorical rather than warranted.

VI. Conclusion

We have presented a controlled simulation study of trust dynamics in 8-player Limit Texas Hold'em. Across four agent architectures, the Pearson correlation between agent trust score and final stack falls along the ladder $-0.752 \to -0.637 \to -0.510 \to +0.062$, with confidence intervals that are wide relative to the inter-phase steps except at the final rung, where the twenty-seed interval ($[-0.10, +0.22]$) is narrower than the step from Phase 3 ($\Delta r_{\text{tp}} = +0.572$). We characterize the rank ordering of the ladder as the robust finding and the absolute size of each step as a preliminary estimate requiring replication. An unbounded hill-climbing sub-experiment with $11\times$ greater parameter drift produces $\Delta r_{\text{tp}} = +0.028$ from bounded Phase 2 and yields a population that diverges rather than converges

in 36-dimensional parameter space— preliminary evidence against the hypothesis that the trap is a bound-box artifact, though Phase 2* varies three optimizer parameters at once and does not cleanly isolate the bounds.

Three follow-up experiments are the most important next steps. *First*, the $n = 20$ Phase 3.1 replication at the 150-hand horizon has now been run and is the dataset reported here; extending it to 500 hands per seed (matching Phase 3) remains the highest-leverage single experiment, as it would remove the hand-count asymmetry from the final rung; estimated cost $\sim$\$230 in API spend. *Second*, completing the scaffold ablation is required before claiming all three components are necessary: the no-CoT arm reported here restores roughly half the trap, but the no-memory and no-notes arms remain to be run. *Third*, a temperature-pinned re-run of Phase 3 at $T = 0.0$ would resolve the open reproducibility caveat (Phase 3.1 is already temperature-pinned). Beyond these three, a multi-LLM replication, a no-limit Hold'em extension, and an experimental varying of inter-archetype distance would each strengthen the present findings.

Why should anyone care outside of cards? All systems where machines assess trust from observed behavior—Credit Scoring, Content Moderation, Multi-Agent Marketplaces, Automated Negotiation—inherit the same threat: if cooperation is apparent, it can be evaluated and exploited. Our results indicate that protection for cooperative agents will likely involve giving them the ability to think about the reputation system itself—rather than simply providing them with additional information about their counterparts. This poker framework serves as a compact, completely reproducible tool for studying the specific question—when does a good reputation become a bad thing?

Competing interests.: The author(s) declare no competing interests.

Data and materials availability.: The full simulation codebase, SQLite databases, figures, tables, and analysis scripts are available at https://anonymous.4open.science/r/trust-sim-anonymous (anonymized repository for review).

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